Code No: 134BD

R16

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year II Semester Examinations, November/December - 2020
FORMAL LANGUAGES AND AUTOMATA THEORY
(Common to CSE, IT)

Time: 2 Hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
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1.a) Draw the transition diagram for a NFA which accepts all the strings with either two consecutive 0's or two consecutive 1's.
b) Define NFA with Epsilon moves. [8+7]

2.a) Write the properties of transition function.
b) Construct DFA accepting the set of all strings with at most one pair of consecutive 0's and at most one pair of consecutive 1's. [7+8]

3.a) Convert regular expression 01*+1 to finite automata.
b) What are the properties of regular expressions? [7+8]

4.a) What are the closure properties of Regular Languages? Explain.
b) Write about the applications of Regular expressions. [7+8]

5. Discuss about:
a) Context Free Grammar
b) Left Most Derivation
c) Right Most Derivation. [5+5+5]

6.a) What is unrestricted grammar? Give an example.
b) Explain the language generated by unrestricted grammar. [7+8]

7. Write the procedure to convert CFG to PDA and also convert the following CFG to PDA.
S → B | aAA
A → aBB | a
B → bBB | A
C → a [15]

8.a) Define the classes P and NP problems. Also write brief note on NP-Complete and NP-Hard Problems.
b) Explain about post correspondence problem. [8+7]

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